

A 39-line arrangement with 470 Kobon triangles

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Abstract. The Kobon triangle problem asks for $K(n)$, the largest number of bounded triangular faces, not crossed by any line, in an arrangement of n straight lines. Before this event no arrangement of 39 lines had been published (OEIS A006066 lists $K(39)$ as “?”); the only upper bound is Tamura’s $\lfloor \frac{n(n-2)}{3} \rfloor = 481$. We give an explicit arrangement of 39 lines with exact integer coefficients and **470** triangles, so $K(39) \geq 470$. It is obtained by adding one line to a published 38-line arrangement with 450 triangles (Parpalak–Utkin gallery); the added line creates 20 triangles and destroys none. The count is checked in exact integer arithmetic by two independent counters and by the contest’s evaluator. We also report an exhaustive sweep (no published 38-line base extends to 471 by one line), several unsuccessful routes beyond 470, first values for other n without published arrangements, and a counting conjecture for even n .

1. Problem and background

An arrangement $\mathcal{A}$ of n lines in $\mathbb{R}^2$ is given by integer triples (a_i, b_i, c_i) , line $a_i x + b_i y + c_i = 0$. A **Kobon triangle** is a bounded face of $\mathcal{A}$ that is a triangle, i.e. a triple of lines whose bounded triangle is crossed in its interior by no other line. Parallel lines and points on three or more lines are allowed. Let $K(n)$ be the maximum count.

Upper bound. Each line carries at most $n - 2$ bounded segments and each triangle uses three of them, so $K(n) \leq \lfloor \frac{n(n-2)}{3} \rfloor$ (Tamura); Felsner and Kriegel [1] prove this for all (also non-simple) arrangements. For $n = 39$ this gives 481, and reaching it forces a *perfect* arrangement (every bounded segment is a side of exactly one triangle) [2]. Bartholdi, Blanc and Loisel [3] and Blanc [4] give sharper bounds for even n in the simple case.

Status of $n = 39$. OEIS A006066 (revision of 14 Sep 2026) lists “?”. The only published exclusion is Savchuk’s SAT result that no perfect 39-line pseudoline arrangement with 13-fold rotational symmetry exists [5]. 39 is not of the form $2m + 1$ with m even, so the doubling construction of [3] does not reach it.

2. Construction

1. **Base.** Arrangement 38-74kq3v76alw of the Parpalak–Utkin gallery [6] (commit cecd2b1): 38 lines, 450 triangles (the best known value for 38), with 5 triple points.
2. **Exact integers.** The gallery stores floating-point slopes and intercepts together with a combinatorial word; we rationalize with the gallery’s own procedure (compact-rationalized, $D = 100$), which keeps the triple points exact and yields coefficients of at most 11 digits, and check the count against the word.
3. **One more line.** For each of 720 directions (fixed random sub-step offset, seed 0) we take a line through the midpoint of every gap between consecutive projections of crossing points, which covers every combinatorially distinct position at that direction. Each candidate is scored incrementally in floating point ($O(n^3)$: surviving triangles plus new triangles through the new line); the best is rounded at scale 10^{12} and re-counted exactly. The added line is

$$490546541635x + 96768230790y - 82521729656546 = 0$$

(slope ≈ -5.0693). It creates 20 new triangles and destroys none of the 450, giving **470**.

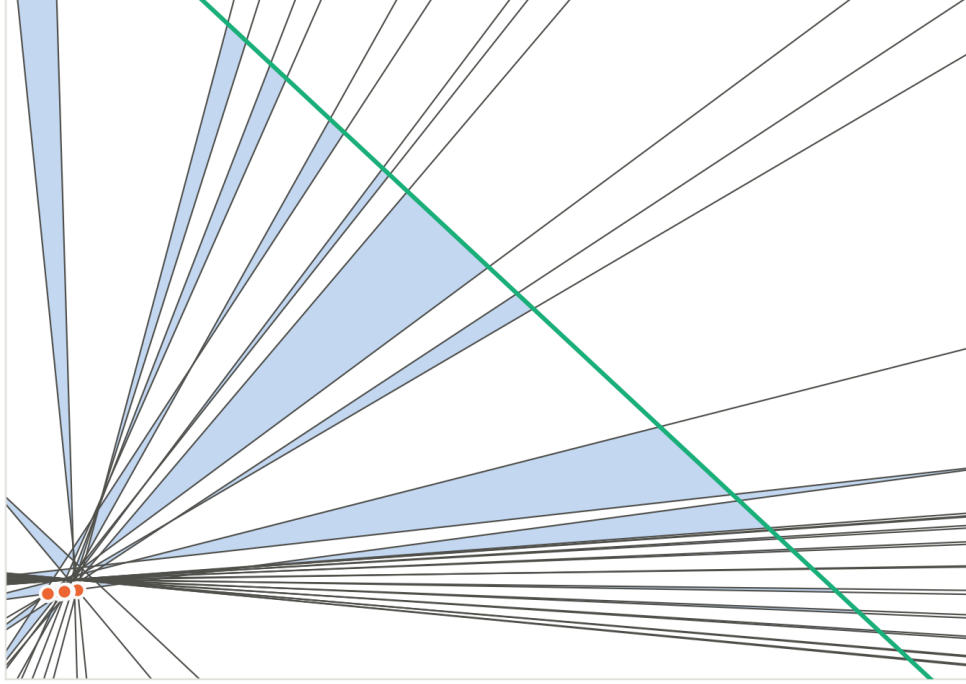


Figure 1: The 39-line arrangement, zoomed on the 20 triangles (shaded) of the added line (green). Orange dots are the five triple points inherited from the base. The full arrangement spans many orders of magnitude.

Sweep. Applying the same step to all 248 gallery arrangements with 38 lines and 450 triangles, with an exhaustive enumeration of the new line’s combinatorial positions, gives the distribution in Fig. 2: 11 bases reach 470 and none reaches 471. Hence **no published 38-line realization extends to 471 by adding one line.**

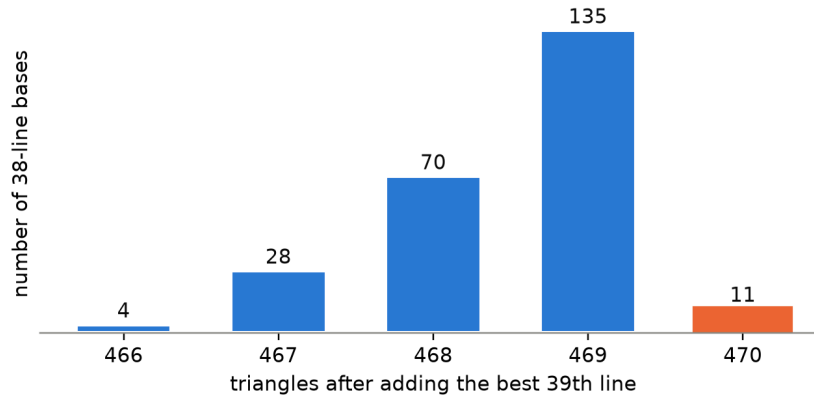


Figure 2: Best count after adding one line, over all 248 published 38-line arrangements with 450 triangles.

3. Verification and checker scope

The solution is the file `kobon-triangles/n39_470/solution.json` (sha256 `85ce027c...3aa352`): 39 distinct integer triples, largest coefficient 14 digits (the contest allows 10^{30}). All intersection points are exact rationals; no floating-point number enters any count.

- **Counter 1** (src/count.py): for every non-degenerate triple, test every other line for vertices strictly on both sides. **Counter 2** (research/tools/verify_independent.py) is an independent implementation. Both give 470.
- **Identity.** $3T = s_1 + 2s_2$ where s_1, s_2 count bounded segments used by one or two triangles: $3 \cdot 470 = 1390 + 2 \cdot 10$. ✓
- **Official evaluation.** The AutoLab hill alejandrozu/kobon-triangles v0.1.0 (tree hash 7d3f1d91dcb8...), with parameter $n = 39$, scored it 470 (passed: true, official: true, 2026-09-27T19:45:26Z); the signed report is included.
- **Reproduction.** sh kobon-triangles/n39_470/reproduce.sh fetches the gallery at the pinned commit, converts the base, adds the line and re-verifies; the output is byte-identical to the submitted file (about 6 s).

Scope. The checkers certify the exact statement “this explicit arrangement has 470 Kobon triangles”, hence $K(39) \geq 470$. They do not address optimality. The trust base is the (short, readable) Python counters and the hill evaluator; no proof assistant was used.

4. Structure

	38-line base	39 lines
triangles	450	470
distinct crossing points	693	731
triple points / parallel pairs	5 / 0	5 / 0
bounded segments E	1353	1428
segments used once / shared by two / unused	1330 / 10 / 13	1390 / 10 / 28
Tamura bound	456	481

Table 1: Segment ledger. Every triple point removes three segments ($E = n(n-2) - 3k$) and here creates two shared segments, so $3T = n(n-2) - k - M$ with M unused segments: $1443 - 5 - 28 = 1410$.

Under this ledger, 471 requires $k + M \leq 30$ (ours has 33) and 481 requires $k = M = 0$, a perfect simple arrangement.

5. Attempts beyond 470

All results below are exact; none produced a straight arrangement above 470.

- Remove one line and re-insert the best one (about 2,900 moves on three 470s; exhaustive single-line replacement on all eleven 470s): the 470s are strict local optima.
- Remove and re-insert 2, 3 or 4 lines (beam search, 1334 / 577 / 82 moves): best 470 (14 further distinct 470s found).
- Force the new line through an existing crossing (new triple point): best 469.
- SAT with 3-fold rotational symmetry (kobon-cnf [5] + Kissat): pseudoline arrangements with 475, 477, 480, 480 and 481 triangles exist — the C3-symmetric perfect case was previously untested — but none could be straightened (best fits violate 339–477 of the 1443 order constraints). The 46 perfect pseudoline words of [7] that we tested also failed to straighten.
- SAT neighbourhood search on a signotope encoding (3–6 free lines, segment budget, exact LP straightening): 394 neighbourhoods, no straight improvement.

6. Partial results

First values for n without published arrangements (exact, reproducible, built by insertion, deletion and doubling from published constructions; most were not officially submitted): $K(40) \geq 494$, $K(44) \geq$

608 (a simple arrangement meeting Blanc’s simple bound), $K(47) \geq 691$, $K(48) \geq 721$, $K(51) \geq 815$, $K(52) \geq 850$, $K(55) \geq 954$, $K(56) \geq 990$, $K(58..61) \geq 1036, 1086, 1137, 1190$, $K(63) \geq 1243$, $K(64) \geq 1302$, $K(66..72) \geq 1329, 1383, 1438, 1494, 1525, 1590, 1657$, and values for 74..96 (repository table). We also give explicit integer realizations (coefficients $< 10^{30}$) of the doubling series for 49, 57, 65, 73, 97 lines, using much larger constants than the proof in [3] requires.

Conjecture (generalized Blanc bound). Let k be the number of triple points, p the number of parallel pairs, and M the number of unused bounded segments. In all 330 arrangements we measured ($n = 6..54$): $E = n(n - 2) - 3k - 2p$ (provable), at most $2k$ segments are shared by two triangles, and, for even n , $k + 2p + M \geq \frac{n}{2} - 1$, i.e. $K(n) \leq \lfloor (n - 2) \frac{2n-1}{6} \rfloor$. This equals the best known value for every even n from 4 to 38. If true, it would give $K(18) = 93$ and settle $K(20) = 117$ for general arrangements; it predicts $K(44) = 609$ (untested). Blanc [4] proves the case $k = p = 0$ with $M \geq \frac{n}{2}$.

7. Novelty and limitations

New: an explicit arrangement with $K(39) \geq 470$ where no value was published; a reproducible pipeline that rebuilds it bit-for-bit from public data; the exhaustive one-line sweep over all published 38-line bases; the existence of C3-symmetric perfect and near-perfect 39-line pseudoline arrangements; first lower bounds for several n ; a counting conjecture with evidence.

Not new or not ours: the 38-line base (Parpalak–Utkin) — 38 of our 39 lines are exactly their arrangement; the doubling construction (Bartholdi–Blanc–Loisel); the SAT encoding (Savchuk); all upper bounds.

Limitations: (i) no optimality claim — the gap to 481 is open, and higher counts for $n = 39$ may be known to others; (ii) the checkers are exact Python programs, not a formal proof-assistant certificate; (iii) the conjecture is unproven and the shared-segment rule is only empirically supported (it is stated informally in [2]); (iv) the “first values” rely on our literature survey for their novelty and are mostly weak relative to the bounds; (v) the reproduction depends on the third-party gallery at a pinned commit, which carries no license and is fetched, not redistributed.

8. Reproducibility and AI disclosure

Requirements: `git`, `uv` (Python 3.12; `numpy`, `numba`, `SciPy`). Commands (repository root): `uv sync`; `uv run python kobon-triangles/n39_470/verify.py`; `sh kobon-triangles/n39_470/reproduce.sh`; `sh kobon-triangles/n39_470/scan_all_bases.sh`. No SAT solver is needed for the 470; Kissat 4.0.4 and `python-sat` were used only for the unsuccessful SAT experiments. The work was carried out with Claude Code (Claude Opus 5.5) as an AI assistant for search, coding and writing; every reported count was verified by exact computation.

Bibliography

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